```sql
create database assignment ;
use assignment ;
CREATE TABLE Employees (
EmpID INT PRIMARY KEY,
Name VARCHAR(50) NOT NULL,
Dept VARCHAR(50) NOT NULL,
City VARCHAR(50) NOT NULL,
Gender CHAR(1) NOT NULL,
Salary INT NOT NULL,
JoinDate DATE NOT NULL,
Role VARCHAR(50) NOT NULL
);

INSERT INTO Employees (EmpID, Name, Dept, City, Gender, Salary, JoinDate, Role) VALUES
(1,'Alice','HR','New York','F',55000,'2019-03-15','Manager'),
(2,'Bob','IT','Chicago','M',72000,'2020-07-22','Developer'),
(3,'Carol','Finance','New York','F',68000,'2018-11-01','Analyst'),
(4,'David','HR','Chicago','M',52000,'2021-01-10','Executive'),
(5,'Eve','IT','Houston','F',85000,'2017-06-30','Senior Dev'),
(6,'Frank','Finance','New York','M',74000,'2019-09-14','Manager'),
(7,'Grace','Marketing','Chicago','F',60000,'2022-02-28','Executive'),
(8,'Hank','IT','Houston','M',90000,'2016-04-05','Lead'),
(9,'Ivy','HR','New York','F',58000,'2020-12-19','Executive'),
(10,'Jack','Finance','Chicago','M',65000,'2021-03-07','Analyst'),
(11,'Karen','Marketing','Houston','F',62000,'2018-08-23','Manager'),
(12,'Leo','IT','New York','M',78000,'2019-11-11','Developer'),
(13,'Mia','HR','Chicago','F',53000,'2022-05-16','Executive'),
(14,'Nate','Finance','Houston','M',71000,'2017-12-02','Manager'),
(15,'Olivia','Marketing','New York','F',66000,'2020-06-18','Analyst'),
(16,'Paul','IT','Chicago','M',82000,'2018-03-27','Senior Dev'),
(17,'Quinn','HR','Houston','F',57000,'2021-09-09','Executive'),
```

```
(18,'Rita','Finance','New York','F',76000,'2019-01-25','Lead'),
(19,'Sam','Marketing','Chicago','M',63000,'2022-07-14','Executive'),
(20,'Tina','IT','Houston','F',88000,'2016-10-31','Lead'),
(21,'Uma','HR','New York','F',60000,'2020-04-03','Manager'),
(22,'Victor','Finance','Chicago','M',69000,'2018-07-19','Analyst'),
(23,'Wendy','Marketing','Houston','F',64000,'2021-11-27','Manager'),
(24,'Xander','IT','New York','M',95000,'2015-02-14','Architect'),
(25,'Yara','HR','Chicago','F',54000,'2022-09-08','Executive'),
(26,'Zoe','Finance','Houston','F',73000,'2019-05-21','Manager'),
(27,'Aaron','Marketing','New York','M',67000,'2020-10-15','Analyst'),
(28,'Bella','IT','Chicago','F',80000,'2017-08-06','Senior Dev'),
(29,'Carlos','HR','Houston','M',56000,'2021-06-24','Executive'),
(30,'Diana','Finance','New York','F',77000,'2018-02-09','Lead');


select * from employees;

-- Queries 1-30
-- Q1. Find the total number of employees in each department
select dept,count(*) as totalemployees from employees group by dept;

-- Q2. Find the total salary paid per department.
select dept, sum(salary) as totalsalary from employees group by dept;

-- Q3. Find the average salary in each city.
select dept, avg(salary) as avgsalary from employees group by dept;

-- Q4. Find the maximum salary in each department.
select dept, max(salary) as maxsalary from employees group by dept;

-- Q5. Find the minimum salary in each department.
```

```
select dept, min(salary) as minsalary from employees group by dept;

-- Q6. List departments with their employee count, sorted by count descending.
SELECT Dept, COUNT(Name) AS Total_Employees FROM employees GROUP BY Dept ORDER BY
COUNT(Name) DESC;

 -- Q7. List cities with total salary, sorted by TotalSalary ascending.
select city , count(salary) as totalsalary from employees group by city;

-- Q8. List all employees ordered by department (A–Z) then by salary (highest first).
SELECT Name, Dept, Salary FROM employees ORDER BY Dept ASC, Salary DESC;

-- Q9. List all employees ordered by city (A–Z) then by join date (oldest first).
select name, city, joindate FROM employees ORDER BY city ASC, joindate DESC;

-- Q10. List all employees ordered by role (A–Z) then by salary (lowest first).
select name, role, salary FROM employees ORDER BY role ASC, salary ASC;

 -- Q11. Show only departments that have more than 7 employees.
select dept, count(*) as empcount from employees group by dept having count(*) >
7;

-- Q12. Show departments where the average salary exceeds 65000.
select dept, avg(salary) as AvgSalary from employees group by dept having avg(salary) > 65000;

-- Q13. Show cities where the total salary bill is greater than 650000.
select city, sum(salary) as totalSalary from employees group by city having sum(salary) > 650000;

-- Q14. Show departments where the maximum salary is at least 80000.
select dept, max(salary) as maxSalary from employees group by dept having max(salary) >= 80000;
```

-- Q15. Show roles that appear more than 5 times in the table.

select role, count(*) as rolecount from employees group by role having count(*) > 5;

-- Q16. Find the overall average salary of all employees

select avg(salary) as companyAvgSalary from employees;

-- Q17. Count how many female employees exist in each department.

select dept, count(Gender) from employees where gender='F' group by dept;

-- Q18. Find the highest and lowest salary across the entire company

select max(salary) as highest, min(salary) from employees;

-- Q19. Find the total salary paid to employees who joined after 2019-01-01.

select sum(salary) as TotalSalary from employees where JoinDate > '2019-01-01';

-- Q20. Find the average salary grouped by gender.

select gender, avg(salary) from employees group by gender;

-- Q21. Find total employees grouped by department AND city.

select dept, city, count(*) from employees group by dept, city;

-- Q22. Find average salary grouped by department AND gender.

select dept, gender, avg(salary) as avgsalary from employees group by dept, gender;

-- Q23. Find total salary grouped by city AND role, ordered by city and total salary descending

select city, role, sum(salary) as totalsalary from employees group by city, role

order by sum(salary) desc;

-- Q24. Count employees grouped by department AND role, show only groups with more than 1 employee.

select dept, role, count(*) from employees group by dept, role having count(*) >

1;

-- Q25. Find max salary grouped by city AND gender.

select city, gender, max(salary) as maxsalary from employees group by city, gender;

-- Q26. Show departments with avg salary > 60000, ordered by avg salary descending.

select dept, avg(salary) as avgaalary from employees group by dept having avg(salary)>60000 order by avg(salary);

-- Q27. Show city+dept combos with more than 1 employee, ordered by count descending then city.

select city, dept, count(*) from employees group by city, dept having count(*)>1

order by count(*) desc, city;

-- Q28. Show roles where total salary exceeds 300000, ordered by total salary ascending.

select role, sum(salary) as totalsalary from employees group by role having sum(salary)>300000 order by sum(salary);

-- Q29. List each employee's name in UPPER CASE along with the year they joined, ordered by join year.

select upper(name) as empname, year(JoinDate) as joinyear from employees;

-- Q30. Find the number of employees who joined each year, sorted by year.

select year(JoinDate) as joinyear, count(*) as joiners from employees group by year(JoinDate) order by year(JoinDate);